

The ACDE Model with Trio and Sibling Data

Contents

1	Introduction	1
2	Quantitative Genetic Model	1
3	Pedigree Covariances	2
3.1	Additive covariance	2
3.2	Dominance covariance	2
3.3	Shared environment	2
3.4	Key pedigree relationships	2
4	Identifiability	2
4.1	Trio data	3
4.2	Trio + sibling data	3
4.3	Summary of identifying equations	3
5	Mixed Model Representation	4
5.1	Vector notation	4
5.2	Phenotypic covariance matrix	4
6	Design Matrices	4
6.1	Additive relationship matrix	4
6.2	Dominance relationship matrix	5
6.3	Shared environment matrix	5
7	R Implementation	5
7.1	Unified model fitting function	5
7.2	Extracting variance components	6
7.3	Model comparison	6
8	Simulation Study	6
8.1	Simulation design	6
8.2	Simulation algorithm	7
8.3	R simulation code	7
8.4	Model recovery experiment	8
8.5	Expected results	8

9 Interpretation	8
10 Conclusion	8

1 Introduction

Classical twin studies partition phenotypic variance into additive genetic (A), shared environmental (C), and unique environmental (E) components. In many settings, however, dominance genetic effects (D) may also contribute substantially to phenotypic variation.

The ACDE model extends the classical ACE framework by including dominance genetic variance. Estimation of all four variance components is not possible with twin data alone because C and D induce similar covariance patterns. Additional pedigree information is therefore required.

This document develops a compact formulation of the ACDE model using trios (mother, father, child) and sibling data. The goal is to derive:

- the quantitative genetic model,
- pedigree covariance expectations,
- identifiability conditions,
- a mixed model representation,
- practical implementation and simulation.

2 Quantitative Genetic Model

For an individual i , the phenotype is written as

$$Y_i = \mu + A_i + D_i + C_i + E_i. \quad (1)$$

The variance components are defined as

$$A_i \sim N(0, \sigma_A^2), \quad (2)$$

$$D_i \sim N(0, \sigma_D^2), \quad (3)$$

$$C_i \sim N(0, \sigma_C^2), \quad (4)$$

$$E_i \sim N(0, \sigma_E^2). \quad (5)$$

The total phenotypic variance is therefore

$$\sigma_P^2 = \sigma_A^2 + \sigma_D^2 + \sigma_C^2 + \sigma_E^2. \quad (6)$$

Additive effects represent the sum of allele substitution effects across loci. Dominance effects represent within-locus interactions between alleles. Shared environment captures envi-

ronmental influences common to members of the same household. Unique environment includes measurement error and individual-specific factors.

3 Pedigree Covariances

Covariances between relatives follow directly from kinship coefficients. Let ϕ_{ij} denote the kinship coefficient between individuals i and j , and let Δ_{ij} denote the probability of sharing both alleles identical by descent at a locus.

3.1 Additive covariance

The additive genetic covariance between individuals i and j is

$$\text{Cov}_A(i, j) = 2\phi_{ij}\sigma_A^2. \quad (7)$$

3.2 Dominance covariance

Dominance covariance depends on the probability of sharing two alleles identical by descent:

$$\text{Cov}_D(i, j) = \Delta_{ij}\sigma_D^2. \quad (8)$$

3.3 Shared environment

Individuals raised in the same household share the same environmental effect:

$$\text{Cov}_C(i, j) = \begin{cases} \sigma_C^2 & \text{if same household,} \\ 0 & \text{otherwise.} \end{cases} \quad (9)$$

3.4 Key pedigree relationships

For common family relationships we obtain:

Relationship	2ϕ	Δ
Parent–child	1/2	0
Full siblings	1/2	1/4
Unrelated	0	0

These coefficients determine all expected covariances used in the ACDE model.

4 Identifiability

A central difficulty of the ACDE model is that shared environment (C) and dominance (D) produce similar covariance structures. Without sufficient pedigree information, the model parameters are not identifiable.

We therefore examine which family structures provide enough information to separate the four variance components.

4.1 Trio data

Consider a nuclear trio consisting of mother (m), father (f), and child (c). Assume parents are unrelated and do not share a household with the child after accounting for the child's shared environment.

The expected covariances are:

$$\text{Var}(Y_c) = \sigma_A^2 + \sigma_D^2 + \sigma_C^2 + \sigma_E^2, \quad (10)$$

$$\text{Var}(Y_m) = \sigma_A^2 + \sigma_D^2 + \sigma_E^2, \quad (11)$$

$$\text{Var}(Y_f) = \sigma_A^2 + \sigma_D^2 + \sigma_E^2. \quad (12)$$

Parent-child covariance:

$$\text{Cov}(Y_c, Y_m) = \frac{1}{2}\sigma_A^2. \quad (13)$$

The same holds for the father.

Key observation Trio data identify the additive variance because parentchild covariance depends only on additive genetic effects. However, trio data alone cannot separate dominance and shared environmental variance because neither produces parentchild covariance.

4.2 Trio + sibling data

Now add a second child (siblings c_1, c_2). Sibling covariance becomes

$$\text{Cov}(Y_{c_1}, Y_{c_2}) = \frac{1}{2}\sigma_A^2 + \frac{1}{4}\sigma_D^2 + \sigma_C^2. \quad (14)$$

This additional equation provides the crucial information required to separate dominance from shared environmental effects.

4.3 Summary of identifying equations

We now have three independent covariance relationships:

$$\text{Parent-child: } \frac{1}{2}\sigma_A^2, \quad (15)$$

$$\text{Sibling: } \frac{1}{2}\sigma_A^2 + \frac{1}{4}\sigma_D^2 + \sigma_C^2, \quad (16)$$

$$\text{Individual variance: } \sigma_A^2 + \sigma_D^2 + \sigma_C^2 + \sigma_E^2. \quad (17)$$

These equations are sufficient to identify all four variance components.

5 Mixed Model Representation

The ACDE model can be written as a linear mixed model.

5.1 Vector notation

Let $\mathbf{y}$ denote the vector of all phenotypes. The model is written as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}_A\mathbf{a} + \mathbf{Z}_D\mathbf{d} + \mathbf{Z}_C\mathbf{c} + \boldsymbol{\varepsilon}. \quad (18)$$

Random effects are assumed to follow:

$$\mathbf{a} \sim N(0, \sigma_A^2 \mathbf{A}), \quad (19)$$

$$\mathbf{d} \sim N(0, \sigma_D^2 \mathbf{D}), \quad (20)$$

$$\mathbf{c} \sim N(0, \sigma_C^2 \mathbf{I}_f), \quad (21)$$

$$\boldsymbol{\varepsilon} \sim N(0, \sigma_E^2 \mathbf{I}). \quad (22)$$

Here:

- $\mathbf{A}$ is the additive relationship matrix,
- $\mathbf{D}$ is the dominance relationship matrix,
- $\mathbf{I}_f$ indexes families.

5.2 Phenotypic covariance matrix

The marginal covariance of $\mathbf{y}$ becomes

$$\mathbf{V} = \sigma_A^2 \mathbf{A} + \sigma_D^2 \mathbf{D} + \sigma_C^2 \mathbf{C} + \sigma_E^2 \mathbf{I}. \quad (23)$$

This form enables estimation via REML using standard mixed-model software.

6 Design Matrices

To implement the ACDE mixed model, we must construct relationship matrices that encode pedigree structure.

6.1 Additive relationship matrix

The additive relationship matrix $\mathbf{A}$ contains twice the kinship coefficients. For the nuclear family with two siblings ordered as

$$(m, f, c_1, c_2),$$

the matrix is

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1/2 & 1/2 \\ 0 & 1 & 1/2 & 1/2 \\ 1/2 & 1/2 & 1 & 1/2 \\ 1/2 & 1/2 & 1/2 & 1 \end{pmatrix}.$$

6.2 Dominance relationship matrix

Dominance relationships capture identity by descent of both alleles. For the same ordering, the dominance matrix becomes

$$\mathbf{D} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1/4 \\ 0 & 0 & 1/4 & 1 \end{pmatrix}.$$

Parents do not share dominance covariance with their children.

6.3 Shared environment matrix

Individuals raised in the same household share a common environmental effect. For a family with two siblings,

$$\mathbf{C} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}.$$

This matrix indexes the family-level random effect.

7 R Implementation

The ACDE model can be estimated using standard mixed model software such as `lme4` or `sommer`. The key requirement is the ability to supply custom covariance matrices.

7.1 Unified model fitting function

Below is a minimal implementation using the `sommer` package.

```
library(sommer)

fit_acde <- function(df, A, D, C) {

  fit <- mmer(
    y ~ 1,
    random = ~ vs(id, Gu = A)      # additive
              + vs(id, Gu = D)      # dominance
              + vs(fam)             # shared environment
    ,
    rcov = ~ units,
    data = df
  )
}
```

```

    return(fit)
}

```

7.2 Extracting variance components

```

extract_vc <- function(fit) {
  vc <- summary(fit)$varcomp
  return(vc[,1])
}

```

7.3 Model comparison

Nested models allow testing of variance components.

```

fit_ADE <- mmer(y ~ 1,
               random = ~ vs(id, Gu=A) + vs(id, Gu=D),
               rcov = ~ units, data=df)

anova(fit_ADE, fit_ACDE)

```

Likelihood ratio tests provide formal evidence for the presence of shared environmental effects.

8 Simulation Study

A simulation study is useful to verify that the ACDE model can recover the true variance components under realistic sampling conditions.

8.1 Simulation design

We simulate nuclear families consisting of:

- mother,
- father,
- two siblings.

True variance components are set to

$$\sigma_A^2 = 0.4, \quad \sigma_D^2 = 0.2, \quad \sigma_C^2 = 0.2, \quad \sigma_E^2 = 0.2.$$

These values produce substantial contributions from all sources of variation.

8.2 Simulation algorithm

For each family j :

1. Draw parental additive values

$$A_m, A_f \sim N(0, \sigma_A^2).$$

2. Generate offspring additive values using Mendelian sampling:

$$A_{c_k} = \frac{1}{2}(A_m + A_f) + M_k, \quad M_k \sim N\left(0, \frac{1}{2}\sigma_A^2\right).$$

3. Simulate dominance deviations:

$$D_i \sim N(0, \sigma_D^2).$$

4. Draw family shared environment:

$$C_j \sim N(0, \sigma_C^2).$$

5. Draw individual environmental noise:

$$E_i \sim N(0, \sigma_E^2).$$

6. Construct phenotypes:

$$Y_i = A_i + D_i + C_j + E_i.$$

8.3 R simulation code

```
simulate_family <- function() {  
  
  Am <- rnorm(1,0,sqrt(.4))  
  Af <- rnorm(1,0,sqrt(.4))  
  
  Cfam <- rnorm(1,0,sqrt(.2))  
  
  make_child <- function(){  
    A <- 0.5*(Am+Af) + rnorm(1,0,sqrt(.2))  
    D <- rnorm(1,0,sqrt(.2))  
    E <- rnorm(1,0,sqrt(.2))  
    return(A + D + Cfam + E)  
  }  
  
  Ym <- Am + rnorm(1,0,sqrt(.2)) + rnorm(1,0,sqrt(.2))  
  Yf <- Af + rnorm(1,0,sqrt(.2)) + rnorm(1,0,sqrt(.2))  
}
```



```

Yc1 <- make_child()
Yc2 <- make_child()

return(c(Ym,Yf,Yc1,Yc2))
}

```

8.4 Model recovery experiment

We simulate many families and fit the ACDE model.

```
Y <- replicate(1000, simulate_family())
```

Estimated variance components should converge to the true values as the sample size increases.

8.5 Expected results

With sufficient sample size, REML estimates are approximately unbiased:

$$\hat{\sigma}_A^2 \approx 0.4, \quad \hat{\sigma}_D^2 \approx 0.2, \quad \hat{\sigma}_C^2 \approx 0.2, \quad \hat{\sigma}_E^2 \approx 0.2.$$

9 Interpretation

The ACDE framework allows separation of additive, dominance, and shared environmental effects using standard family data.

Key insights include:

- Parent–child covariance identifies additive genetic variance.
- Sibling covariance separates dominance from shared environment.
- Mixed models provide a convenient estimation framework.

The approach extends classical twin models while retaining a simple likelihood-based implementation.

10 Conclusion

Trio and sibling data provide sufficient information to estimate the full ACDE variance decomposition. The mixed model formulation allows practical implementation using existing software and enables flexible extensions to larger pedigrees and genomic relationship matrices.